Math F252 Final Exam Fall 2025

Name: Solutions

Rules:

You have 120 minutes to complete this midterm.

Partial credit will be awarded, but you must show your work.

Calculators are not allowed.

One hand-written sheet of notes is allowed.

Turn off anything that might go beep during the exam.

Good luck!

Problem	Possible	Score
1	15	
2	6	
3	6	
4	10	
5	20	
6	6	
7	10	
8	10	
9	6	
10	5	
11	6	
Extra Credit	5	
Total	100	

1. (5 pts each) Evaluate the indefinite integrals.

(a) $\int \frac{1}{x^2\sqrt{x^2-4}}dx$ $= \int \frac{2\sec\theta\tan\theta\,d\theta}{4\sec^2\theta \cdot 2\tan\theta} = \frac{1}{4}\int \frac{1}{\sec\theta}\,d\theta$

trig substitution

let $x = 2\sec\theta$

$dx = 2\sec\theta\tan\theta\,d\theta$

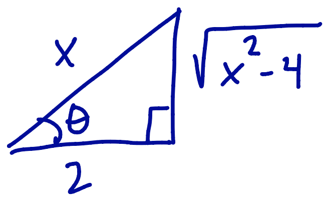
$\sqrt{x^2-4} = \sqrt{4\tan^2\theta}$

$= 2\tan\theta$

$= \frac{1}{4}\int \cos\theta\,d\theta = \frac{1}{4}\sin\theta + C$

$= \frac{1}{4}\frac{\sqrt{x^2-4}}{x} + C$

$\frac{x}{2} = \sec\theta$



(b) $\int \frac{1}{x^2-9}dx$ $= \int \frac{1/6}{x-3}dx + \int \frac{-1/6}{x+3}dx$

partial fractions

$= \frac{1}{6}\ln|x-3| - \frac{1}{6}\ln|x+3| + C$

$\frac{1}{x^2-9} = \frac{A}{x+3} + \frac{B}{x-3}$

$= \frac{1}{6}\ln\left|\frac{x-3}{x+3}\right| + C$

$1 = A(x-3) + B(x+3)$

let $x=3$: $1 = 6B$.

So $B = 1/6$

$x=-3$: $1 = -6A$

So $A = -1/6$

(c) $\int (4x - xe^x)dx = \int 4x\,dx - \boxed{\int xe^x dx}$

$= 2x^2 - (xe^x - e^x) + C = 2x^2 - xe^x + e^x + C$

aside IBP

$\boxed{\int xe^x dx} = xe^x - \int e^x dx = xe^x - e^x$

$u = x \quad dv = e^x dx$

$du = dx \quad v = e^x$

2. (6 pts) The shaft of a bird feather has density function $\boxed{\rho = \frac{2}{5}\arctan x}$ grams per meter on the interval from $x = 0$ m to $x = 1$ m. Find the **mass** of the shaft. **Include units with your answer.**

$\text{mass} = \int_0^1 \frac{2}{5}\arctan x\,dx = \frac{2}{5}\left[x\arctan x\Big|_0^1 - \int_0^1 \frac{x}{1+x^2}dx\right]$

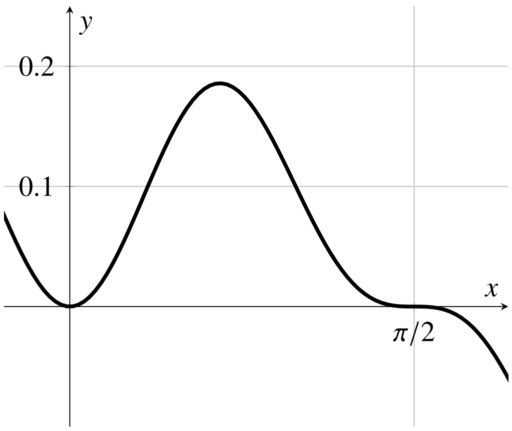
IBP

$\boxed{\begin{matrix} u = \arctan x & dv = dx \\ du = \frac{1}{1+x^2}dx & v = x \end{matrix}}$

$= \frac{2}{5}(\arctan(1)) - \frac{2}{5}\cdot\frac{1}{2}\cdot\ln(1+x^2)\Big|_0^1$

$= \frac{2}{5}\left(\frac{\pi}{4}\right) - \frac{1}{5}(\ln(2) - \ln(1)) = \boxed{\frac{\pi}{10} - \frac{1}{5}\ln(2)}$ grams

3. (6 pts) The region R is bounded by $y = \sin^2(x)\cos^3(x)$ and the x-axis between $x = 0$ and $x = \pi/2$. Find the **area** of the region R.



$$A = \int_0^{\pi/2} \sin^2(x)\cos^3(x)\,dx$$

$$= \int_0^{\pi/2} (1-\sin^2 x)\sin^2 x \cos(x)\,dx$$

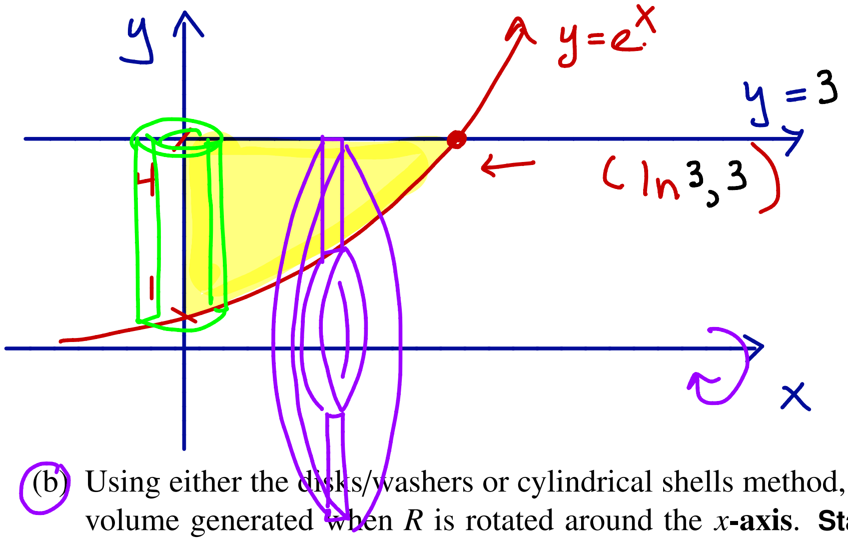
Let $u = \sin(x)$
$du = \cos(x)\,dx$
If $x=0$, $u=0$.
If $x=\pi/2$, $u=1$.

$$= \int_0^1 (1-u^2)u^2\,du = \int_0^1 (u^2-u^4)\,du$$

$$= \frac{1}{3}u^3 - \frac{1}{5}u^5 \Big|_0^1 = \frac{1}{3}-\frac{1}{5} = \frac{2}{15}$$

4. (10 pts) The region R is bounded by $y = e^x$, $y = 3$, and the y-axis.

   (a) Sketch the region R. Label the curves and all points of intersection of the curves.



   (b) Using either the disks/washers or cylindrical shells method, **set up an integral** to compute the volume generated when R is rotated around the x**-axis**. **State which method you are using.** You do not need to evaluate the integral.

washers

$$V = \pi \int_0^{\ln 3} (3)^2 - (e^x)^2 \, dx = \pi \int_0^{\ln 3} (9 - e^{2x}) \, dx$$

   (c) Using either the disks/washers or cylindrical shells method, **set up an integral** to compute find the volume generated when A is rotated around the y**-axis. State which method you are using.** You do not need to evaluate the integral.

Cylindrical shells

$$V = 2\pi \int_0^{\ln 3} x(3 - e^x) \, dx = 2\pi \int_0^{\ln 3} (3x - xe^x) \, dx$$

5. (5 pts each) Determine whether each series below converges or diverges. Name the test you use and justify your conclusion. (This means show your work!)

(a) $\sum_{n=1}^{\infty} \cos\left(\frac{1}{5n^2+1}\right)$ Test: divergence Converge or Diverge

$$\lim_{n\to\infty} \cos\left(\frac{1}{5n^2+1}\right) = \cos(0) = 1 \neq 0$$

(b) $\sum_{n=1}^{\infty} \frac{2+3n}{3+4n^{3/2}}$ Test: limit comparison Converge or Diverge

Compare to $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$, a divergent p-series

$$\lim_{n\to\infty} \left(\frac{(2+3n)}{3+4n^{3/2}}\right) \cdot \frac{\sqrt{n}}{1} = \lim_{n\to\infty} \frac{2\sqrt{n}+3n^{3/2}}{3+4n^{3/2}} = \frac{3}{4}$$ ← a number

(c) $\sum_{n=1}^{\infty} \frac{2^n}{n^3}$ Test: ratio Converge or Diverge

$$\lim_{n\to\infty} \frac{2^{n+1}}{(n+1)^3} \cdot \frac{n^3}{2^n} = \lim_{n\to\infty} 2\left(\frac{n^3}{(n+1)^3}\right) = 2 > 1$$

(d) $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{3n+2}}$ Test: alternating series Converge or Diverge

$$b_n = \frac{1}{\sqrt{3n+2}}$$

① $\lim_{n\to\infty} \frac{1}{\sqrt{3n+2}} = 0$ ✓

② $b_{n+1} = \frac{1}{\sqrt{3n+5}} < \frac{1}{\sqrt{3n+2}} = b_n$ ✓

6. (6 pts) Use the Integral Test to determine whether the series $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n\sqrt{\ln(n)}}$ converges or diverges.

$$\int_2^{\infty} \frac{(\ln(x))^{-1/2}}{x}\,dx = \lim_{t\to\infty} \int_2^{t} \frac{(\ln(x))^{-1/2}}{x}\,dx$$

$$= \lim_{t\to\infty} 2(\ln x)^{\frac{1}{2}} \Big|_2^t = \lim_{t\to\infty} \left(2\sqrt{\ln t} - 2\sqrt{\ln(2)}\right)$$

$$= \infty$$

So the series diverges because the integral diverges.

7. (5 pts each) For each power series below, determine the interval of convergence.

(a) $\displaystyle\sum_{n=1}^{\infty} \frac{(x-3)^n}{n5^n}$

$$\lim_{n\to\infty} \left| \frac{(x-3)^{n+1}}{(n+1)5^{n+1}} \cdot \frac{n5^n}{(x-3)^n} \right| = \frac{|x-3|}{5}\left(\lim_{n\to\infty} \frac{n}{n+1}\right) = \frac{|x-3|}{5}$$

We want $\frac{|x-3|}{5} < 1$. So $-5 < x-3 < 5$ or $-2 < x < 8$

Check endpoints

$x=-2$: $\sum \frac{(-5)^n}{n5^n} = \sum \frac{(-1)^n}{n}$, alt. harmonic

$x=8$: $\sum \frac{5^n}{n5^n} = \sum \frac{1}{n}$, harmonic

Answer

$[-2, 8)$

(b) $\displaystyle\sum_{n=1}^{\infty} \frac{n!(x+4)^n}{7^{2n}}$

$$\lim_{n\to\infty} \left| \frac{(n+1)!(x+4)^{n+1}}{4^{2n+2}} \cdot \frac{4^{2n}}{n!(x+4)^n} \right| = \lim_{n\to\infty} |x+4| \left(\frac{n+1}{16}\right) = \infty$$

So the series converges only at $x=-4$.

So the interval of convergence is $[-4,-4]$.

8. (10 pts) Recall that the Maclaurin series $\cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$.

   (a) Write $p_4(x)$, the 4th-degree Maclaurin **polynomial** for $\cos(x)$.

$$P_4(x) = 1 - \frac{x^2}{2} + \frac{x^4}{24}$$

   (b) Find the Maclaurin series for $f(x) = x\cos(2x)$. Simplify your answer.

$$x\cos(2x) = x\sum_{n=0}^{\infty} \frac{(-1)^n (2x)^{2n}}{n!} = \sum_{n=0}^{\infty} \frac{(-1)^n 2^{2n} x^{2n+1}}{n!}$$

   (c) Find the Maclaurin series for $G(x) = \int_0^x \cos(\sqrt{t})\, dt$.

$$G(x) = \int_0^x \left(\sum_{n=0}^{\infty} \frac{(-1)^n t^n}{(2n)!}\right) dt = \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+1}}{(n+1)(2n)!}$$

9. (6 pts) Consider the curve $x(t) = t^2 - 3t + 1$, $y(t) = e^t$.

   (a) Determine the slope of the curve at the point $(1, 1)$.

We need the time t so that

$x(t) = t^2 + 3t + 1 = 1$ and

$y(t) = e^t = 0$. So $t = 0$.

Now

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{e^t}{2t-3}; \quad \text{So } \left.\frac{dy}{dx}\right|_{t=0} = \frac{e^0}{2 \cdot 0 - 3} = \frac{1}{-3} = \frac{-1}{3}$$

   (b) Determine the points where the tangent line is horizontal or vertical, or state that none exist.

horizontal: Need $\frac{dy}{dt} = e^t = 0$. Never.

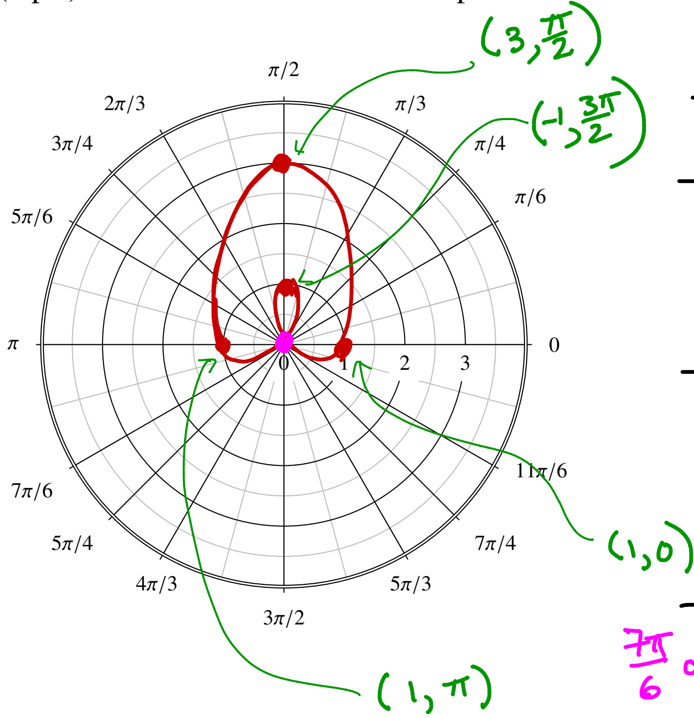
vertical: Need $\frac{dx}{dt} = 2t - 3 = 0$. So $t = 3/2$.

$x(3/2) = (3/2)^2 - 3\left(\frac{3}{2}\right) + 1 = \frac{9}{4} - \frac{9}{2} + 1 = \frac{-5}{4}$

$y(3/2) = e^{3/2}$

point $\left(\frac{-5}{4}, e^{3/2}\right)$

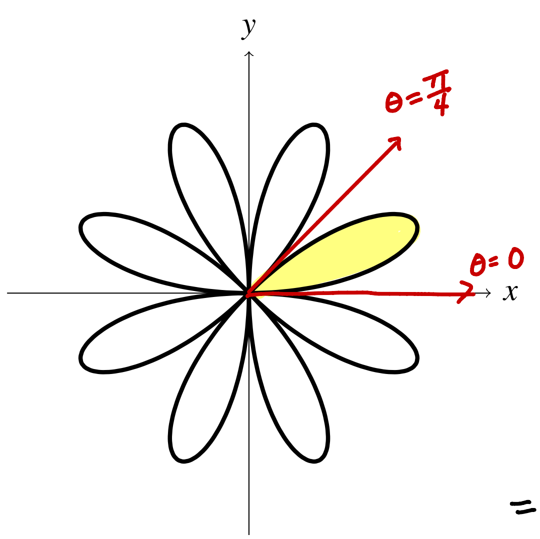
10. (5 pts) Make a careful sketch of the polar curve $r = 1 + 2\sin(\theta)$. $\boxed{\text{Label at least 4 points.}}$



θ	$r = 1+2\sin\theta$
0	1
$\frac{\pi}{2}$	3
π	1
$\frac{3\pi}{2}$	-1
$\frac{7\pi}{6}$ or $\frac{11\pi}{6}$	0 ←

$$0 = 1 + 2\sin\theta;\quad \sin\theta = \frac{-1}{2};\quad \theta = \frac{7\pi}{6} \text{ or } \frac{11\pi}{6}$$

11. (6 pts) Compute the area enclosed by the polar curve $r = 3\sin(4\theta)$. The graph of the curve is shown below.



$$A = 8\int_0^{\pi/4} \frac{1}{2}\left(3\sin(4\theta)\right)^2 d\theta$$

$$= 36\int_0^{\pi/4} \sin^2(4\theta)\, d\theta$$

$$= 18\int_0^{\pi/4} \left(1 - \cos(8\theta)\right) d\theta$$

$$= 18\left(\theta - \frac{1}{8}\sin(8\theta)\right)\Big|_0^{\pi/4}$$

$$= 18\left(\left(\frac{\pi}{4} - \frac{1}{8}\sin(2\pi)\right) - (0 - \sin 0)\right) = \frac{18\pi}{4} = \frac{9}{2}\pi$$

Extra Credit (5 points) Let $f(x) = \sum_{n=0}^{\infty} \binom{\frac{1}{2}}{n} \frac{x^n}{3^n}$.

(a) Determine the first 4 terms in the series. Simplify the coefficients.

$$\binom{\frac{1}{2}}{0} + \binom{\frac{1}{2}}{1}\frac{x}{3} + \binom{\frac{1}{2}}{2}\frac{x^2}{3^2} + \binom{\frac{1}{2}}{3}\frac{x^3}{3^3}$$

$$= 1 + \frac{1}{2} \cdot \frac{x}{3} + \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)}{2!}\frac{x^2}{9} + \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{3!} \cdot \frac{x^3}{27}$$

$$= 1 + \frac{1}{6}x - \frac{1}{36}x^2 + \frac{1}{216}x^3$$

(b) The series converges when $x = 2$. Determine the sum.

$$\sum_{n=0}^{\infty} \binom{\frac{1}{2}}{n}\left(\frac{x}{3}\right)^n = \left(1 + \frac{x}{3}\right)^{\frac{1}{2}}$$ where it converges.

So $$\sum_{n=0}^{\infty} \binom{\frac{1}{2}}{n}\left(\frac{2}{3}\right)^n = \left(1 + \frac{2}{3}\right)^{\frac{1}{2}} = \sqrt{\frac{5}{3}}.$$